

II. Related Work — Draft v3

Draft for the GS³LAM semantic-safety navigation paper (target: ICRA 2027). v3, revised 2026-07-19: seven previously missing papers verified and integrated, AlphaAdj citation resolved (arXiv:2603.21142), Ames et al. CBF citation added, VISA cut pending a real citation. Every citation below was verified in GS3LAM_PAPER_SCOPE.md/ARCHITECTURE.md or independently confirmed against arXiv/SPIE this session (authors, IDs, and descriptions checked against the actual abstracts). One [TODO] remains — see the flag list at the end.

Our work sits at the intersection of four bodies of literature: Gaussian Splatting as a map representation for robots, semantic scene understanding in navigation, Control Barrier Functions as safety filters, and the use of Vision-Language Models as a perceptual reasoning component. We review each in turn and position our contribution — a VLM-derived, per-object safety scalar grafted onto the Gaussian map and used to weight a collision-cone CBF — against the closest work in each area.

A. 3D Gaussian Splatting for Robotics

3D Gaussian Splatting (3DGS) represents a scene as an explicit set of anisotropic Gaussian primitives, a form that is simultaneously renderable, differentiable, and — because each primitive has an explicit position and covariance — directly usable by downstream planning and control modules. This explicitness is what our pipeline exploits: a splat can carry an extra scalar attribute, and a controller can read it.

GS³LAM [Li et al., MM '24] is the SLAM front end our system builds on. It runs semantic segmentation (DEVA), camera tracking, and map optimization concurrently, folding a semantic term into the mapping loss alongside the photometric and geometric terms, so that splat boundaries come to respect object boundaries during reconstruction rather than through any post-hoc label projection. The result is a Gaussian map in which every splat carries a semantic identity vector. GS³LAM's semantics, however, stop at identity: the map records *which* object a splat belongs to, and nothing about what that object implies for a robot moving near it. Our contribution begins exactly where GS³LAM's ends — we consume its semantically grouped map and attach a safety judgment to each object. We emphasize that GS³LAM is prior work we build upon, not a component we claim.

GS³LAM is part of a broader semantic Gaussian SLAM family. SGS-SLAM [Li et al., ECCV '24, arXiv:2402.03246] was the first semantic visual SLAM system built on Gaussian Splatting, optimizing appearance, geometry, and semantics as parallel channels with semantic-guided keyframe selection; GSFF-SLAM [Lu et al., arXiv:2504.19409] instead decouples the semantic field from the geometric reconstruction. Both share GS³LAM's scope and its limitation for our purposes: the output is a labeled map, with no notion of hazard and no connection to control.

Splat-Nav [Chen et al., arXiv:2403.02751] navigates directly through Gaussian maps, constructing safe-by-construction polytope corridors through the splat field and localizing the robot against it in real time. Splatblox [Chopra et al., arXiv:2511.18525] targets outdoor navigation, fusing segmented RGB and LiDAR into a Gaussian representation to build a traversability-aware Euclidean signed-distance field that distinguishes traversable vegetation (tall grass) from rigid obstacles (tree trunks) over missions up to 100 m. Gau-Occ [Lv et al., arXiv:2603.22852] predicts 3D semantic occupancy for autonomous driving from compact semantic Gaussians, initialized by diffusion-based LiDAR completion and fused with multi-view image features. Across all three, the semantics in play are navigability classes — free, occupied, traversable — and both objects of equal occupancy constrain the robot equally. Our system answers a complementary

question: among the places the robot *can* go, how cautiously should it treat each object it passes, given what that object is. NavGSim [Liu et al., arXiv:2603.15186], finally, uses hierarchical 3DGS to build photorealistic floor-scale *simulators* for navigation, slicing navigable areas directly from the Gaussians to train and evaluate policies; it uses the splat field to make training environments, where we use it as the live substrate of a control-time safety constraint.

A separate cluster hardens Gaussian SLAM against dynamic scenes. WildGS-SLAM [Zheng et al., CVPR '25, arXiv:2504.03886] uses uncertainty-aware mapping to identify and remove moving distractors from monocular reconstruction; DL-SLAM [Xu et al., arXiv:2607.01860] couples pixel-level and object-level dynamic probabilities to prune dynamic Gaussians while letting transiently static objects aid tracking; DynaSplaTAM [Zhang, Proc. SPIE 14244, ICCIA 2026, DOI 10.1117/12.3113817] achieves lightweight dynamic-object masking with a YOLOv8-plus-depth module, reducing trajectory error relative to SplaTAM on TUM RGB-D. These are map-integrity mechanisms, not semantic ones — orthogonal to hazard reasoning, and complementary to our pipeline, which currently assumes a static scene; DL-SLAM's per-object dynamic probability in particular is the kind of backbone our future dynamic-scene extension would build on.

B. Semantic Scene Understanding and VLMs in Navigation

A useful axis for organizing semantic navigation work is *labeling versus reasoning*. Systems like GS³LAM, SGS-SLAM, and GSFF-SLAM label: they assign each region a categorical identity. A second family embeds language features into the 3D representation itself. GaussianVLM [Halacheva et al., arXiv:2507.00886] is the strongest recent representative: it associates linguistic features with each Gaussian primitive and sparsifies them into task-aware tokens for embodied question answering and scene description. Reasoning of this kind is latent and textual — retrieval and generation over scene tokens [TODO: LangSplat-family citations for the earlier open-vocabulary retrieval line — Flag 2]. It does not produce a physical-hazard quantity, and nothing in a similarity score encodes that a glass vase near a doorway is a hazard to a wheeled robot while a concrete pillar is not — that conclusion is not a nearest-neighbor lookup but an inference about material, fragility, and consequence. Our use of a generative VLM is aimed at precisely this gap: the model is asked to render a safety judgment about an object, not to embed it for later matching. This distinction between similarity-based semantics and judgment-based semantics is, we believe, the load-bearing one for safety applications.

The cost of invoking a VLM online has independently produced two-tier "fast-and-slow" navigation architectures. FSR-VLN [Zhou et al., arXiv:2509.13733] matches against a hierarchical multi-modal scene graph first and escalates to VLM reasoning only to refine goal selection, reporting an 82% reduction in response time relative to VLM-based baselines; HiCo-Nav [Xu et al., arXiv:2604.21363] decouples perception, memory, and reasoning into asynchronous layers so that VLM inference never blocks real-time control. We share their premise — generative VLM calls are too expensive for the inner loop — but resolve it differently: rather than escalating online, our hero-frame mechanism moves every VLM call offline and amortizes it to one query per object. We regard the cheap-first-expensive-later principle as established practice these systems have validated, not as a contribution of ours; what is ours is applying it to the synthesis of control constraints.

The closest prior use of VLM judgment in navigation is AlphaAdj [Chen and Chandra, arXiv:2603.21142], which queries a VLM at a fixed cadence (2 Hz, asynchronously) over the robot's egocentric RGB stream for a bounded scalar risk estimate of the current view, and maps that estimate onto the class-K gain of a CBF safety filter — managing roughly 0.7 s of query latency with staleness gating and a speed-aware cap. We note the closeness candidly: modulating a CBF's class-K gain with VLM-assessed risk is the same lever as one of our

two weighting modes. The differences define our contribution. AlphaAdj produces one scene-level scalar per view, transient and image-bound, that adjusts the filter's *global* conservativeness, and it must keep paying VLM latency online. We produce one judgment *per object*, persisted as an attribute of the 3D map, so each splat's constraint is weighted individually — a fragile vase and the wall behind it tighten the barrier by different amounts within the same view — and the control loop incurs zero VLM latency because every query happened offline against the object's hero frame.

C. Control Barrier Functions for Robot Navigation

Control Barrier Functions provide a minimally invasive safety layer: a nominal controller proposes a command, and a quadratic program modifies it as little as possible subject to constraints that keep the system inside a safe set [Ames et al., ECC 2019]. The relevant question for map-based navigation is how the safe set is constructed from the environment representation, and a recent line of work builds it directly from Gaussian splats.

SAFER-Splat [Chen et al., arXiv:2409.09868] formulates a distance-based CBF per splat for safe navigation with online Gaussian Splatting maps. Its constraint is purely geometric: every splat repels the robot according to proximity, and two objects at equal distance generate equal constraints regardless of what they are. Our system differs on exactly this point — the per-splat safety scalar makes constraint strength depend on the object's semantics, not only its geometry.

Khass et al. [arXiv:2605.20566] extend splat-based CBFs toward uncertainty-awareness: a barrier derived from an Average Value-at-Risk collision metric over the splats' covariances is combined with perception barrier functions in a unified QP, so the robot trades information gathering against collision risk while safety remains a hard constraint. The uncertainty being priced, however, is geometric — the map's own confidence. A perfectly reconstructed hazard and a perfectly reconstructed sofa constrain the robot identically; semantic differentiation is precisely the layer this line of work has not addressed.

Tscholl, Nakka, and Gunter [arXiv:2509.14421] is the geometric foundation of our Stage 3, and we want to be precise about the relationship: it is the primitive we build on, not a competing approach. Their contribution is analytic. Treating each splat as a confidence ellipsoid, they derive a closed-form collision cone in relative-velocity space, obtaining a first-order barrier function per splat that slots into a QP filter without high-order CBF machinery, with robot geometry handled by inflating the ellipsoid bound. On a scene of roughly 170,000 splats they report a threefold reduction in planning time and lower trajectory jerk than distance-based per-splat barriers at a comparable level of safety. Their formulation is entirely geometric — the paper contains no semantic component of any kind — and that is precisely why it composes cleanly with our work. We adopt their collision-cone CBF unchanged as the geometric substrate, and contribute the layer they do not address: per-splat semantic safety scores that modulate the constraint, either through the class-K gain (scaling how aggressively the barrier brakes for an object) or through the splat's effective covariance (widening the region the cone routes around). The two contributions are orthogonal — theirs is how a splat constrains motion; ours is how much, as a function of what the splat is.

The same authors have since extended the primitive along the dynamics axis: FastBridge [Tscholl, Nakka, and Gunter, arXiv:2607.01200] replaces the reduced-order model with the full quadrotor dynamics, deriving a high-relative-degree exponential collision-cone CBF plus a backup CBF that preserves QP feasibility under actuator limits, reporting 47% lower trajectory jerk at $2.25\times$ the speed of a state-of-the-art 3DGS safety filter. The extension is, again, entirely geometric — which reinforces the point: the collision-cone substrate is

actively evolving on the dynamics side while its semantic side remains open, and our weighting layer is in principle composable with such refinements.

D. VLMs for Robot Perception

A broader line of work applies VLMs to robot perception and safety outside the map-building loop, and it contains the works closest to ours.

The closest in spirit is Semantic-Metric Bayesian Risk Fields [Chen et al., arXiv:2512.08233], which learns human-aligned, continuous risk from demonstration videos: a pretrained VLM furnishes a Bayesian prior over risk, a learned ViT likelihood aligns it with human preference, and the output is a pixel-dense, query-conditioned 2D risk image, projected into 3D as a cost for trajectory optimization or a value signal for planners. The differences sit exactly where our contribution lives. Their risk exists in image space and is projected per use; ours is a persistent, first-class attribute of the 3D map. Their semantic signal is a learned likelihood over VLM features; ours is an explicit reasoned judgment queried per object. And their downstream consumer is a trajectory optimizer's cost; ours is a hard CBF constraint in the control loop.

The closest on the representation side is PhysGS [Chopra et al., CVPR '26, arXiv:2511.18570], which grafts VLM-derived quantities onto splats: material labels, density estimates, and confidence scores are fused by Bayesian posterior refinement across views into dense per-splat beliefs over friction, hardness, stiffness, and mass. What it grafts, however, is physical property estimation built from categorical material priors — not a reasoned judgment about hazard — and the resulting field feeds no controller. Relatedly, Phys2Real [Wang et al., arXiv:2510.11689, ICRA '26] shows that VLM-inferred priors over physical parameters such as center of mass, fused with interactive adaptation, can condition RL manipulation policies for sim-to-real transfer. Both demonstrate that VLMs carry usable physical knowledge; neither converts that knowledge into navigation-time safety constraints, which is the conversion this paper is about.

ASIMOV [Sermanet et al., CoRL '25] approaches semantic safety at the level of behavior: it generates robot "constitutions" and benchmarks for evaluating whether robot conduct is semantically safe. It does not attach safety information to a spatial representation or to a controller, which is where our work operates. The two are complementary rather than overlapping — we view ASIMOV-style benchmarks as a candidate source of ground truth for validating whether our VLM-produced safety scores are themselves accurate, an evaluation our experimental protocol anticipates.

A final family attacks VLM latency directly. EAMP [Huang et al., arXiv:2606.25629] distills VLM outputs into a small local model so that online queries need no API call; AsyncVLA [Hirose et al., arXiv:2602.13476] instead runs the large model remotely as asynchronous high-level guidance while a lightweight onboard adapter refines actions at high frequency, tolerating multi-second communication delays. Our design sidesteps the same problem structurally rather than by compression or asynchrony — hero-frame querying moves all VLM calls offline and amortizes them to one per object, so the online system incurs zero VLM latency by construction, and the consumer of the semantic signal is a certified constraint rather than a learned policy. Distillation remains attractive as future work for the setting our current design does not cover: previously unseen objects appearing at runtime, where an offline precomputation cannot help.

GWM [Lu et al., ICCV '25, arXiv:2508.17600] develops Gaussian World Models, extending Gaussian representations from reconstruction toward prediction of future scene states. Our safety field describes the scene as currently mapped; a predictive layer of this kind is the natural direction for anticipatory safety — reacting to a hazard before it materializes — and we discuss it as future work rather than claiming it here.

Positioning

The pieces of our system all now exist separately in the verified literature: analytic geometric CBFs on splats (Tscholl et al.; FastBridge; SAFER-Splat), uncertainty-aware splat CBFs (Khass et al.), VLM-derived physical properties grafted onto splats (PhysGS), VLM-conditioned continuous risk for robot motion (Bayesian Risk Fields), and online VLM risk estimates regulating a CBF's gain (AlphaAdj). The Stanford Multi-Robot Systems Lab is the dominant adjacent group, publishing along two tracks — geometric splat-based control (Splat-Nav, SAFER-Splat) and VLM-derived physics and risk (Phys2Real, Bayesian Risk Fields) — that its own publications have not yet connected. We therefore do not claim that using a VLM to assess physical hazard for robot control is new; it is not. The claim of this paper is the specific unclosed loop: an explicit, reasoned VLM safety judgment rendered once per object via a hero-frame query, stored as a persistent first-class attribute of the Gaussian map, and consumed as a per-splat weight on an analytic collision-cone CBF at control time. To our knowledge, no published system performs that synthesis, and we scope the contribution accordingly.

Flags for Aashrut — judgment calls, gaps, and errata (not part of the paper)

Flag 1 — Citation status: one TODO left. All seven previously missing papers are now verified and integrated (NavGSim 2603.15186, Gau-Occ 2603.22852, WildGS-SLAM 2504.03886, SGS-SLAM 2402.03246, DL-SLAM 2607.01860, FastBridge 2607.01200, AsyncVLA 2602.13476). AlphaAdj is resolved (Flag 3), Ames et al. is cited (Flag 4), VISA is cut (Flag 5). The only remaining TODO is the LangSplat-family citation in Section B — GaussianVLM carries most of that weight, but the original open-vocabulary retrieval line (LangSplat/LERF) should still be cited once sourced.

Flag 2 — Deep-research-report errata (verified false or wrong, kept on record): (a) the report attributed the collision-cone paper to *Chen et al.*, *IEEE T-RO Jan 2025* — false; it is Tscholl, Nakka & Gunter, arXiv:2509.14421 (preprint for IEEE L-CSS/ACC). (b) DynaSplat's author is Yujing Zhang, not "Chen et al." (c) The report's spelled-out COVER title is a wrong backronym (real title: "Coverage Optimization for Camera View Selection").

Flag 3 — AlphaAdj: resolved, and upgraded in importance. Found via targeted search: Chen & Chandra, arXiv:2603.21142 — the system is named AlphaAdj in the abstract. Verified details correct ARCHITECTURE.md §4's description: it queries at 2 Hz asynchronously (not frame-by-frame) with staleness gating around ~695 ms latency, and it maps VLM risk onto the CBF's class-K gain — the same lever as your ALPHA_SCALE mode. Recommend updating the ARCHITECTURE.md §4 table row accordingly. The rewritten Section B paragraph concedes the closeness explicitly and differentiates on per-object vs. scene-level, persistent-3D vs. transient-2D, and offline vs. online-latency grounds; it is now one of the three most load-bearing paragraphs (with Bayesian Risk Fields and Tscholl).

Flag 4 — Resolved. Section C now opens with Ames, Coogan, Egerstedt, Notomista, Sreenath & Tabuada, "Control Barrier Functions: Theory and Applications," ECC 2019, pp. 3420–3431 (sourced from the Spring 2025 URP PDF's reference list, per Aashrut) — the standard citation for exactly this framing.

Flag 5 — VISA: cut, per two failed targeted searches. No arXiv record matches "VISA (Xian et al.)" under any plausible expansion; nearby hits (Vi-LAD arXiv:2503.09820, VITA-VLA arXiv:2510.09607) are different papers by different authors. What's lost: only a second example in the distillation sentence — EAMP alone carries the point, and AsyncVLA now fills the latency-architecture space. Reinstate only if your lit review doc yields a real citation.

Flag 6 — Real but excluded papers (unchanged). LEGS (2606.01458), COVER (2604.05259), GaussLite (2606.30809) verified real but off the safety/semantics/control axis.

Flag 7 — Remaining venue caveats. PhysGS's CVPR 2026 is now cited per your confirmation. WildGS-SLAM is cited as CVPR 2025 on your twice-confirmed verification — the arXiv abstract page itself doesn't display the venue tag, so double-check the camera-ready citation format. Splat-Nav's "TRO 2024" (report's claim) remains unverified; cited as arXiv. FSR-VLN's 82% figure is confirmed verbatim in its abstract, with scope "compared to VLM-based methods on tour videos" — the draft now says "relative to VLM-based baselines" to match.

Flag 8 — Judgment call: naming the Stanford MSL explicitly (unchanged from v2) — tone-check with Dr. Yel.

Flag 9 — Judgment call: concessive framing of two-tier architectures (unchanged from v2) — trim if too concessive.

Flag 10 — The three defensive paragraphs. Bayesian Risk Fields, AlphaAdj, and the Tscholl orthogonality paragraph now jointly carry the novelty claim. Each difference stated in them is verified against the actual abstracts, but these are the paragraphs a skeptical reviewer reads first — review word by word.

Flag 11 — Unchanged from v2: GSFF-SLAM's description is still one verified line deep; GWM placement (Area D vs. Future Work) remains your call.